

FIELD THEORY

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1. DEFINITIONS

1.1. Algebraic field extensions.

Definition 1.1.1 (Field). A *field* is commutative division ring. In other words, a field is a commutative ring for which all nonzero elements have a multiplicative inverse.

Definition 1.1.2. Let K and L be fields (Definition 1.1.1).

1. A map $\phi : K \rightarrow L$ is a *field homomorphism* if it satisfies the following axioms for all $x, y \in K$:
 - (a) $\phi(x + y) = \phi(x) + \phi(y)$ (additivity);
 - (b) $\phi(xy) = \phi(x)\phi(y)$ (multiplicativity);
 - (c) $\phi(1_K) = 1_L$ (unitality).

Equivalently, a field homomorphism is a ring homomorphism (Definition A.0.7) between fields.

If such a map exists, we often say that *K embeds into L* ; this terminology is justified because field homomorphisms are injective (Definition A.0.8) as set maps (Proposition 1.1.3).

2. A field homomorphism $\phi : K \rightarrow L$ is an *isomorphism* if it is bijective (Definition A.0.8).
3. The set of all homomorphisms from a field K to a field L is denoted by $\text{Hom}(K, L)$.
4. Assuming that K and L have a common subfield F , an *F -embedding of K into L* is an embedding $K \rightarrow L$ that acts as the identity map (Definition A.0.9) on F . The set of F -embeddings of K into L is denoted by $\text{Hom}_F(K, L)$.

If K and L are fields such that K embeds into L , then that means that there is a subfield (Definition 1.1.5) of L that is isomorphic to K .

Proposition 1.1.3. Let K and L be fields (Definition 1.1.1) and let $\phi : K \rightarrow L$ be a field homomorphism (Definition 1.1.2). Then ϕ is injective (Definition A.0.8).

Definition 1.1.4 (Vector space over a field). Let $(k, +, \cdot)$ be a field (Definition 1.1.1). A *vector space over k* or a *k -vector space* is a triple $(V, +, \cdot)$ ¹ where

1. $(V, +)$ is an abelian group, and
2. $\cdot$ is a map $k \times V \rightarrow V$, called *scalar multiplication*

such that the following axioms hold for all $a, b \in k$ and all $u, v \in V$:

1. (Compatibility with field multiplication)

$$(ab) \cdot v = a \cdot (b \cdot v).$$

2. (Identity scalar)

$$1 \cdot v = v.$$

3. (Distributivity over vector addition)

$$a \cdot (u + v) = a \cdot u + a \cdot v.$$

4. (Distributivity over scalar addition)

$$(a + b) \cdot v = a \cdot v + b \cdot v.$$

Definition 1.1.5 (Field Extension). Let K be a field and let L be a field such that $K \subseteq L$ and the operations of K are the restrictions of those of L . Then L is called a *extension field* (or just an *extension*) of K . The notation L/K is often used synonymously; we say that L/K is a *field extension*. Moreover, K is said to be a *subfield of L* .

Definition 1.1.6. Let R be a ring. There exists a unique ring homomorphism (Definition A.0.7) $\psi : \mathbb{Z} \rightarrow R$ defined by mapping $n \mapsto n \cdot 1_K$. The non-negative generator of the kernel of this map, $\ker(\psi) = \langle p \rangle \subseteq \mathbb{Z}$, is called the *characteristic of R* , denoted by $\text{char}(R)$.

Proposition 1.1.7. Let K be a field (Definition 1.1.1). Then $\text{char}(K)$ (Definition 1.1.6) is either 0 or a prime number p .

Definition 1.1.8. (♠ TODO: finite field, rational numbers) Let K be a field. The *prime subfield of K* is the intersection of all subfields (Definition 1.1.5) of K .

- If $\text{char}(K) = p > 0$, the prime subfield is isomorphic to the finite field $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$.
- If $\text{char}(K) = 0$, the prime subfield is isomorphic to the field of rational numbers $\mathbb{Q}$.

Lemma 1.1.9. Let L/K be a field extension (Definition 1.1.5). Then L is a vector space (Definition 1.1.4) over K .

Definition 1.1.10 (Degree of an Extension, Finite Extension). (♠ TODO: dimension) Let L/K be a field extension (Definition 1.1.5). The *degree* of the extension $[L : K] := \dim_K(L)$, i.e. the dimension of L as a K -vector space (Lemma 1.1.9). If $[L : K] < \infty$, the extension L/K is called a *finite field extension* and L is said to be a *finite extension of K* .

¹Note that $+$ and $\cdot$ are abuse of notation here as these are already used for the addition and multiplication of $\cdot$.

Definition 1.1.11 (Algebraic Element, Algebraic Extension). Let L/K be a field extension (Definition 1.1.5) and let $x \in L$.

- If there exists a nonzero polynomial $f(t) \in K[t]$ such that $f(x) = 0$, then x is called an *algebraic element over K* . There exists a unique such monic irreducible polynomial $f(t)$, which is called the *minimal polynomial of x over K* .
- Otherwise, x is called a *transcendental element over K* .

If every $x \in L$ is algebraic over K , then L/K is called an *algebraic extension*.

Definition 1.1.12. A field (Definition 1.1.1) F is said to be *algebraically closed* if every nonconstant polynomial $f(x) \in F[x]$ has a root in F , i.e., for every such $f(x)$ there exists $a \in F$ with $f(a) = 0$.

Definition 1.1.13. Let K be a field. A field extension L/K is called an *algebraic closure of K* if the following two conditions hold:

1. Every element $a \in L$ is algebraic over K (Definition 1.1.11).
2. The field L is algebraically closed (Definition 1.1.12).

One often writes an algebraic closure of K by $\overline{K}$.

Definition 1.1.14 (Field Generated by Elements). Let L/K be a field extension (Definition 1.1.5) and let $S \subseteq L$ be a subset. The *field generated by S over K* is the smallest subfield of L that contains K and S . Equivalently, it is the intersection of all subfields of L containing $K \cup S$. This field is denoted by $K(S)$.

In the special case where $S = \{x_1, \dots, x_n\}$ is finite, one writes $K(x_1, \dots, x_n)$. If $n = 1$, then $K(x)$ is called a *simple field extension*.

Definition 1.1.15 (Finitely generated field extension). Let E/F be a field extension (Definition 1.1.5). We say that E/F is *finitely generated* if there exist finitely many elements $\alpha_1, \dots, \alpha_n \in E$ such that $E = F(\alpha_1, \dots, \alpha_n)$ (Definition 1.1.14), where $F(\alpha_1, \dots, \alpha_n)$ denotes the smallest subfield of E containing F and $\{\alpha_1, \dots, \alpha_n\}$.

Definition 1.1.16 (Field Automorphism). Let L be a field. A *field automorphism of L* is a bijective ring homomorphism $L \rightarrow L$.

Definition 1.1.17 (Automorphism Group of a Field Extension). Let L/K be a field extension (Definition 1.1.5).

- An automorphism (Definition 1.1.16) $\sigma : L \rightarrow L$ is called a *K -automorphism* if σ is a field automorphism of L that fixes every element of K , i.e. $\sigma(a) = a$ for all $a \in K$.
- The set of all such K -automorphisms of L forms a group (Definition A.0.2) under composition, called the *automorphism group of the extension L/K* . It is usually denoted by $\text{Aut}(L/K)$ or $\text{Aut}_K(L)$.

Definition 1.1.18 (Normal Extension). Let K be a field and let $S \subseteq K[t]$ be a set of polynomials.

- A field $L \supseteq K$ is a *splitting field of S over K* if

- every polynomial in S splits completely into linear factors over L , and
- L is generated over K (Definition 1.1.14) by the roots of the polynomials in S , namely

$$L = K(\{\alpha \mid f(\alpha) = 0, f \in S\}).$$

- An algebraic extension L/K is called a **normal extension** if L is the splitting field of a family of polynomials in $K[t]$.

Definition 1.1.19 (Separable Element, Separable Extension). Let L/K be a field extension (Definition 1.1.5) and let $x \in L$ be algebraic over K (Definition 1.1.11) with minimal polynomial (Definition 1.1.11) $m_{x,K}(t) \in K[t]$.

- The element x is **separable over K** if $m_{x,K}(t)$ has distinct roots in a splitting field (Definition 1.1.18).
- The element x is **inseparable over K** otherwise.

An algebraic extension (Definition 1.1.11) L/K is called **separable extension** if every element $x \in L$ is separable over K .

See also Definition 1.2.6, which defines separable field extensions in greater generality.

Definition 1.1.20. A field F is said to be **separably closed** if every nonconstant separable polynomial $f(x) \in F[x]$ has a root in F , i.e., for every such $f(x)$ there exists $a \in F$ with $f(a) = 0$ (equivalently, every separable polynomial factors completely into linear factors over F).

Definition 1.1.21. Let K be a field (Definition 1.1.1). A field extension (Definition 1.1.5) L/K is called a **separable closure of K** if the following hold:

1. The extension L/K is separable algebraic (Definition 1.1.19) (i.e., every $a \in L$ is separable over K).
2. The field L is separably closed (Definition 1.1.20).

One often writes a separable closure of K by K^{sep} . Some also write separable closures of K by $\overline{K}$, which may conflict with the common notation for algebraic closures of K .

Definition 1.1.22 (Galois Extension). An extension (Definition 1.1.5) L/K is called a **Galois extension** if it is both a normal extension (Definition 1.1.18) and a separable algebraic extension (Definition 1.1.19). Its **Galois group**, usually denoted by $\text{Gal}(L/K)$, is defined to be the automorphism group $\text{Aut}(L/K)$ (Definition 1.1.17).

Theorem 1.1.23 (Fundamental Theorem of Galois Theory). Let L/K be a finite (Definition 1.1.10) Galois extension of fields (Definition 1.1.22) with Galois group $G = \text{Gal}(L/K)$. Then there is an inclusion-reversing bijection between the set of subgroups $H \leq G$ and the set of intermediate fields $K \subseteq F \subseteq L$, given by (♠ TODO: L^H)

$$H \longmapsto L^H := \{x \in L \mid \sigma(x) = x \text{ for all } \sigma \in H\},$$

$$F \longmapsto \text{Gal}(L/F).$$

This correspondence has the following properties:

- (i) L^H is a subfield of L containing K for each subgroup $H \leq G$.
- (ii) $\text{Gal}(L/F)$ is a subgroup of G for each intermediate field F .
- (iii) $H_1 \subseteq H_2 \iff L^{H_2} \subseteq L^{H_1}$.
- (iv) $[L^H : K] = |G : H|$, $[L : L^H] = |H|$.
- (v) F/K is Galois if and only if $\text{Gal}(L/F) \trianglelefteq G$,
in which case $\text{Gal}(F/K) \cong G / \text{Gal}(L/F)$.

Corollary 1.1.24 (Normality of the whole extension). If L/K is a finite (Definition 1.1.10) Galois extension of fields (Definition 1.1.22) with Galois group G , then $K = L^G$. (♠ TODO: L^G)

Theorem 1.1.25 (Primitive element theorem). Let L/K be a finite (Definition 1.1.10) separable extension (Definition 1.1.19). Then there exists an element $\alpha \in L$ such that

$$L = K(\alpha).$$

(Definition 1.1.14) In other words, every finite separable extension is simple (Definition 1.1.14).

Lemma 1.1.26 (Normal extension characterization). (♠ TODO: embedding) Let L/K be a finite extension. Then L/K is normal (Definition 1.1.18) if and only if every K -embedding $\sigma : L \rightarrow \overline{K}$ into an algebraic closure (Definition 1.1.11) $\overline{K}$ satisfies $\sigma(L) = L$.

Theorem 1.1.27 (Infinite Galois Theory). Let L/K be a (possibly infinite) Galois extension with Galois group $G = \text{Gal}(L/K)$. Endow G with the Krull topology, i.e. the unique compact, Hausdorff, totally disconnected topology whose neighborhood basis at the identity is given by the open subgroups $\text{Gal}(L/F)$, where F runs through the finite Galois intermediate extensions $K \subseteq F \subseteq L$.

Then there is an inclusion-reversing bijection between: - the closed subgroups $H \leq G$, and - the intermediate fields $K \subseteq F \subseteq L$,

given by

$$\begin{aligned} H &\longmapsto L^H := \{x \in L \mid \sigma(x) = x \text{ for all } \sigma \in H\}, \\ F &\longmapsto \text{Gal}(L/F). \end{aligned}$$

This correspondence satisfies:

- (i) $H \leq G$ closed $\iff F = L^H$ for some intermediate field F .
- (ii) F/K is finite Galois $\iff \text{Gal}(L/F)$ is open in G .
- (iii) F/K is Galois $\iff \text{Gal}(L/F) \trianglelefteq G$,
in which case $\text{Gal}(F/K) \cong G / \text{Gal}(L/F)$.

Proposition 1.1.28 (Krull topology characterization). For a (possibly infinite) Galois extension L/K , the Galois group $G = \text{Gal}(L/K)$ is a profinite group, i.e. a compact, Hausdorff, totally disconnected topological group isomorphic to an inverse limit of finite groups.

Definition 1.1.29. A *field* k is said to be a *perfect field* if it satisfies one of the following equivalent conditions: (♠ TODO: irred polynomial, root)

1. Every irreducible polynomial over k has distinct roots in an algebraic closure (Definition 1.1.11) of k (i.e., every algebraic extension (Definition 1.1.11) of k is separable (Definition 1.1.19)).
2. The characteristic of k is 0, or the characteristic of k is $p > 0$ and the Frobenius endomorphism $\Phi : k \rightarrow k$, defined by $\Phi(x) = x^p$, is an automorphism.

Proposition 1.1.30. Let k be a field (Definition 1.1.1). If k is algebraically closed (Definition 1.1.12), of characteristic (Definition 1.1.6) 0, or a finite field, then k is a perfect field (Definition 1.1.29).

1.2. Transcendence degrees.

Definition 1.2.1 (Algebraically independent elements). Let E/F be a field extension (Definition 1.1.5), and let $S = \{x_i\}_{i \in I}$ be a subset of E indexed by some set I . We say that S is *algebraically independent over F* if for every finite subset $\{x_{i_1}, \dots, x_{i_n}\} \subseteq S$ and every non-zero polynomial $P \in F[X_1, \dots, X_n]$, we have $P(x_{i_1}, \dots, x_{i_n}) \neq 0$. Equivalently, S is algebraically independent over F if the natural ring homomorphism (Definition A.0.7)

$$F[X_i : i \in I] \rightarrow E$$

sending $X_i \mapsto x_i$ is injective, where $F[X_i : i \in I]$ denotes the polynomial ring (Definition A.0.4) over F in the indeterminates $\{X_i\}_{i \in I}$.

Definition 1.2.2 (Transcendence basis). Let E/F be a field extension (Definition 1.1.5). A subset $T \subseteq E$ is called a *transcendence basis* of E over F if:

1. T is algebraically independent over F , i.e., there is no non-trivial polynomial relation with coefficients in F among finitely many elements of T , and
2. E is algebraic (Definition 1.1.11) over $F(T)$, where $F(T)$ denotes the field generated by (Definition 1.1.14) F and T .

Definition 1.2.3 (Transcendence degree). Let E/F be a field extension (Definition 1.1.5). The *transcendence degree of E over F* , denoted $\text{tr. deg}_F(E)$ or $\text{trdeg}_F(E)$, is defined as the cardinality (Definition A.0.5) of any transcendence basis (Definition 1.2.2) of E over F . By a classical result, all transcendence bases of E over F have the same cardinality, so the transcendence degree is well-defined.

If E/F is algebraic (Definition 1.1.11), then $\text{tr. deg}_F(E) = 0$. If no transcendence basis is finite, then $\text{tr. deg}_F(E)$ is an infinite cardinal.

Definition 1.2.4 (Function field). Let k be a field (Definition 1.1.1). A field K is called a *function field over k* (or a *function field in n variables over k*) if K is a finitely generated field extension (Definition 1.1.15) of k with transcendence degree (Definition 1.2.3) n over k , where $n \geq 1$. Equivalently, the function field in n variables over k is isomorphic to the fraction field (Definition A.0.6) of the polynomial ring (Definition A.0.4) $k[x_1, \dots, x_n]$ where $x_1, \dots, x_n$ are indeterminate variables. Accordingly, the function field in n variables over k is often denoted as $k(x_1, \dots, x_n)$.

When $n = 1$, we say that K is a *function field of one variable over k* or an *algebraic function field over k* and denote it by $k(x)$.

Definition 1.2.5 (Separating transcendence basis). Let E/F be a field extension (Definition 1.1.5), and let $T \subseteq E$ be a transcendence basis of E over F . We say that T is a **separating transcendence basis** if the algebraic extension $E/F(T)$ (Definition 1.1.14) is separable (Definition 1.1.19), i.e., every element of E is separable over $F(T)$ (meaning that its minimal polynomial (Definition 1.1.11) over $F(T)$ has no repeated roots in an algebraic closure).

Definition 1.2.6 (Separable field extension). Let E/F be a field extension (Definition 1.1.5). In general (allowing transcendental extensions), E/F is called **separable** if there exists a separating transcendence basis (Definition 1.2.5) for E over F . Equivalently, E/F is separable if every finitely generated (Definition 1.1.15) intermediate field $F \subseteq K \subseteq E$ has a separating transcendence basis over F . In particular, a field extension E/F is algebraic and separable in the above sense if and only if it is a separable algebraic extension in the sense of Definition 1.1.19.

APPENDIX A. MISCELLANEOUS DEFINITIONS

Definition A.0.1. Let X and Y be sets. A **map** (or **function**) from X to Y is a rule f assigning to each element $x \in X$ exactly one element $f(x) \in Y$. We write $f : X \rightarrow Y$.

We say that X is the **domain** and that Y is the **codomain of f** .

Definition A.0.2 (Groups). A **group** is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a binary operation, subject to the following conditions:

1. (Associativity) For all $g, h, k \in G$ one has

$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$

2. (Identity element) There exists an element $e \in G$ such that for all $g \in G$,

$$e \cdot g = g \cdot e = g.$$

3. (Inverse element) For all $g \in G$ there exists an element $g^{-1} \in G$ such that

$$g \cdot g^{-1} = g^{-1} \cdot g = e.$$

The element e is called the **identity element of G** , and g^{-1} is called the **inverse of g** .

Equivalently, a group is a monoid with inverse elements.

A group $(G, \cdot)$ is often simply written as G , when the notation for the binary operation $\cdot$ is clear.

An **abelian group** or synonymously, a **commutative group**, is a group $(G, \cdot)$ whose binary operation $\cdot$ is **abelian** or **commutative**, i.e. satisfies

$$g \cdot h = h \cdot g$$

for all $g, h \in G$.

Definition A.0.3. Let $(R, +, \cdot)$ and $(S, +, \cdot)$ be rings, not assumed to be commutative. A function $f : R \rightarrow S$ is called a **ring homomorphism** if for all $r_1, r_2 \in R$ the following properties hold:

1. $f(r_1 + r_2) = f(r_1) + f(r_2)$,
2. $f(r_1 r_2) = f(r_1) f(r_2)$,
3. $f(1_R) = 1_S$ where 1_R and 1_S denote the multiplicative identities in R and S , respectively.

A ring homomorphism is said to be a **ring isomorphism** if it is invertible as a map of sets.

An **R -ring** refers to a ring S equipped with a ring homomorphism $f : R \rightarrow S$.

We note that a ring homomorphism $f : R \rightarrow S$ yields a natural left R -module structure on S and a natural right R -module structure on S respectively as follows for $r \in R$ and $s \in S$:

$$r \cdot s = f(r) \cdot s$$

$$s \cdot r = s \cdot f(r).$$

However, these left and right module structures need not yield a two-sided R -module structure.

Definition A.0.4 (Polynomial ring over a commutative ring). Let R be a commutative ring. For a set of variables $\{x_i\}_{i \in I}$, the **polynomial ring in variables $\{x_i\}$ over R** , denoted by notations such as $R[x_i \mid i \in I]$ or $R[x_i]_{i \in I}$, is defined as the commutative R -algebra whose elements are finite R -linear combinations of monomials in the variables x_i , where the variables commute with each other and with elements of R .

That is, $R[x_i \mid i \in I]$ is the free commutative R -algebra generated by the set $\{x_i\}$.

In the case that I is a finite set, and writing $y_1, \dots, y_n$ for the variables x_i , it is customary to let $R[y_1, \dots, y_n]$ denote the polynomial ring.

Definition A.0.5. An ordinal number κ is a **cardinal number** (or simply a **cardinal**) if for every ordinal $\alpha < \kappa$, there is no bijection (Definition A.0.8) between α and κ . Equivalently, a cardinal is an initial ordinal—an ordinal that is not equinumerous with any smaller ordinal.

The **cardinality of an arbitrary set X** , denoted by $|X|$, $\text{card}(X)$, or $\#X$, is the unique cardinal number κ such that there exists a bijection between X and κ . (The existence of such a κ for every set requires the Axiom of Choice).

Definition A.0.6. Let R be an , and consider the set $R \times (R \setminus \{0\})$ as above. Define a relation $\sim$ on $R \times (R \setminus \{0\})$ by declaring that

$$(a, b) \sim (c, d) \quad \text{if and only if} \quad ad = bc,$$

for $a, c \in R$ and $b, d \in R \setminus \{0\}$. This relation is an equivalence relation. Its equivalence classes are denoted by $\frac{a}{b}$.

The set of equivalence classes

$$\left\{ \frac{a}{b} \mid a \in R, b \in R \setminus \{0\} \right\}$$

under the relation $\sim$ defined above is called the **field of fractions of R** , and is denoted by $\text{Frac}(R)$.

The operations on $\text{Frac}(R)$ are defined by

$$\begin{aligned}\frac{a}{b} + \frac{c}{d} &= \frac{ad+bc}{bd}, \\ \frac{a}{b} \cdot \frac{c}{d} &= \frac{ac}{bd},\end{aligned}$$

for $a, c \in R$ and $b, d \in R \setminus \{0\}$. With these operations, $\text{Frac}(R)$ is a field (Definition 1.1.1).

Definition A.0.7. Let $(R, +, \cdot)$ and $(S, +, \cdot)$ be rings, not assumed to be commutative. A function $f : R \rightarrow S$ is called a **ring homomorphism** if for all $r_1, r_2 \in R$ the following properties hold:

1. $f(r_1 + r_2) = f(r_1) + f(r_2)$,
2. $f(r_1 r_2) = f(r_1) f(r_2)$,
3. $f(1_R) = 1_S$ where 1_R and 1_S denote the multiplicative identities in R and S , respectively.

A ring homomorphism is said to be a **ring isomorphism** if it is invertible as a map of sets.

An **R -ring** refers to a ring S equipped with a ring homomorphism $f : R \rightarrow S$.

We note that a ring homomorphism $f : R \rightarrow S$ yields a natural left R -module structure on S and a natural right R -module structure on S respectively as follows for $r \in R$ and $s \in S$:

$$r \cdot s = f(r) \cdot s$$

$$s \cdot r = s \cdot f(r).$$

However, these left and right module structures need not yield a two-sided R -module structure.

Definition A.0.8. Let X and Y be sets and let $f : X \rightarrow Y$ be a function.

- The function f is said to be **injective** (or **one-to-one**) if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- The function f is said to be **surjective** (or **onto**) if for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- The map f is **bijective** if it is both injective and surjective. In this case, there exists a unique **inverse map** $f^{-1} : Y \rightarrow X$ such that for all $x \in X$ and $y \in Y$,

$$f^{-1}(f(x)) = x \text{ and } f(f^{-1}(y)) = y.$$

Definition A.0.9. Let X be a set. The **identity function on X** , denoted by id_X , is the function (Definition A.0.1) $\text{id}_X : X \rightarrow X$ defined by

$$\text{id}_X(x) = x \text{ for all } x \in X.$$

It is the unique function on X satisfying $f \circ \text{id}_X = f = \text{id}_X \circ f$ for every function $f : X \rightarrow Y$ and every function $f : Y \rightarrow X$.

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